

$$\begin{aligned}
C_{\phi\text{ud}}^{(11)} i1-i2 \rightarrow & \frac{1}{16\pi^2} \left(-\frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{2,1,1,0} [m_1, m_d^{-i2}, m_q^{\text{p}}] + \frac{1}{36} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \right. \\
& \overline{y_u^{\text{pi}1}} \text{LF}_{2,2,1,-1} [m_1, m_d^{-i2}, m_q^{\text{p}}] + \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{3,1,1,-1} [m_1, m_d^{-i2}, m_q^{\text{p}}] - \\
& \frac{1}{36} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{2,2,1,-1} [m_1, m_q^{\text{p}}, m_d^{-i2}] - \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{2,1,1,0} [m_1, m_q^{\text{p}}, \tilde{\mu}] + \\
& \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{3,1,1,-1} [m_1, m_q^{\text{p}}, \tilde{\mu}] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_d^{-i2}] - \\
& \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^{\text{p}}] - \frac{3}{2} m_2 \tilde{\mu} g_2^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{2,1,1,0} [m_2, m_q^{\text{p}}, \tilde{\mu}] + \\
& \frac{3}{2} m_2 \tilde{\mu} g_2^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{3,1,1,-1} [m_2, m_q^{\text{p}}, \tilde{\mu}] + \frac{3}{2} m_2 \tilde{\mu} g_2^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{\text{p}}] + \\
& \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{2,1,1,0} [m_3, m_d^{-i2}, m_q^{\text{p}}] - \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{2,2,1,-1} [m_3, m_d^{-i2}, m_q^{\text{p}}] - \\
& \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{3,1,1,-1} [m_3, m_d^{-i2}, m_q^{\text{p}}] + \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{2,2,1,-1} [m_3, m_q^{\text{p}}, m_d^{-i2}] - \\
& \frac{1}{4} \tilde{\mu} y_d^{\text{pi}2} \overline{y_u^{\text{si}1}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,2,1,-1} [m_q^{\text{p}}, \tilde{\mu}, m_u^{\text{r}}] + \\
& \frac{1}{4} \tilde{\mu} y_d^{\text{pi}2} \overline{y_u^{\text{si}1}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,2,1,-1} [m_u^{\text{r}}, \tilde{\mu}, m_q^{\text{p}}] - \frac{1}{3} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_1, m_d^{-i2}] + \\
& \frac{1}{3} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_1, m_d^{-i2}] + \frac{1}{2} \tilde{\mu} y_d^{\text{pi}2} \overline{y_u^{\text{si}1}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_q^{\text{p}}, m_u^{\text{r}}] - \\
& \frac{1}{2} \tilde{\mu} y_d^{\text{pi}2} \overline{y_u^{\text{si}1}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_q^{\text{p}}, m_u^{\text{r}}] - \\
& \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{1,1,1,1,0} [m_1, m_d^{-i2}, m_q^{\text{p}}, \tilde{\mu}] - \frac{1}{12} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \\
& \text{LF}_{2,1,1,1,-1} [m_1, m_d^{-i2}, m_q^{\text{p}}, \tilde{\mu}] - \frac{1}{3} \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{a_u^{\text{pi}1}} \text{LF}_{1,1,1,1,0} [m_1, m_q^{\text{p}}, m_u^{\text{i}1}, \tilde{\mu}] + \\
& \frac{1}{6} \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{a_u^{\text{pi}1}} \text{LF}_{2,1,1,1,-1} [m_1, m_q^{\text{p}}, m_u^{\text{i}1}, \tilde{\mu}] - \frac{1}{12} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \\
& \text{LF}_{2,1,1,1,-1} [m_d^{-i2}, m_1, m_q^{\text{p}}, \tilde{\mu}] - \frac{2}{9} \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{a_u^{\text{pi}1}} \text{LF}_{2,1,1,1,-1} [m_q^{\text{p}}, m_1, m_d^{-i2}, m_u^{\text{i}1}] + \\
& \frac{1}{12} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \text{LF}_{2,1,1,1,-1} [m_q^{\text{p}}, m_1, m_d^{-i2}, \tilde{\mu}] - \frac{1}{6} \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{a_u^{\text{pi}1}} \\
& \text{LF}_{2,1,1,1,-1} [m_q^{\text{p}}, m_1, m_u^{\text{i}1}, \tilde{\mu}] + \frac{4}{3} \tilde{\mu} g_3^2 y_d^{\text{pi}2} \overline{a_u^{\text{pi}1}} \text{LF}_{2,1,1,1,-1} [m_q^{\text{p}}, m_3, m_d^{-i2}, m_u^{\text{i}1}] + \\
& \frac{1}{6} \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{a_u^{\text{pi}1}} \text{LF}_{2,1,1,1,-1} [m_u^{\text{i}1}, m_1, m_q^{\text{p}}, \tilde{\mu}] + \frac{1}{12} m_1 \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{y_u^{\text{pi}1}} \\
& \text{LF}_{2,1,1,1,-1} [\tilde{\mu}, m_1, m_d^{-i2}, m_q^{\text{p}}] - \frac{1}{6} \tilde{\mu} g_1^2 y_d^{\text{pi}2} \overline{a_u^{\text{pi}1}} \text{LF}_{2,1,1,1,-1} [\tilde{\mu}, m_1, m_q^{\text{p}}, m_u^{\text{i}1}] \Big)
\end{aligned}$$